



Title: Data Science Languages and Tools. Assignment

Please go through all the requirements for this assignment and read them carefully.

PURPOSE OF THE ASSIGNMENT:

Assess the ability of each student to use the python programming languages to train and evaluate classification models.

TASK OF THE ASSIGNMENT:

Start from the notebook we used through this module in topic 1 (R and Python) and 3 (Libraries for Data Science).

I ask you to focus on Hyperparameters optimization. More in detail, I would ask you to work on:

- "max_depth"
- "min_samples_leaf"

What I ask you to do is:

- Changing such Hyperparameters values
- Training a model
- Measuring the effect on performance

Some hints about evaluating results

- Choose a performance evaluation setup. One example: precision over the 3000 customers with the highest score, recall. You can choose yours.
- Assess such performance on the "dtc3" model we trained during the lectures. This will be your "baseline performance" benchmark.
- Assess the performance on the new models you trained.
- Try to explain why you got a better/worse performance.

It is not important to achieve better performance. What is important is to prove your ability to perform experiment and measure results mindfully.

Final deliverable

Describe, in a very brief document (no more than two pages), the findings you obtained and the most important code snippets you had to write. The ability to stay within the limits will be evaluated.

TECHNICAL REQUIREMENTS:

You can use google colab notebook we built in our lectures. In case you prefer to use some other python environment, that is fine.

GRADING CRITERIA:

1. Ability to change the code and run it
2. Ability to explain and comment the results.



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ADDITIONAL SOURCES TO BE USED:

Anything you can find on the internet. Do not use “people”! Be mindful about the usage of AI.

REFERENCING:

The lectures’ slide decks and notebooks.

PLAGIARISM DISCLAIMER (NOTE FOR THE STUDENT): “I hereby declare that this assignment is my own work and all references created are as per the Harvard Referencing System”.